
BCA191 - PROJECT MANUAL



H023

RIKISHI-H023

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INTRODUCTION

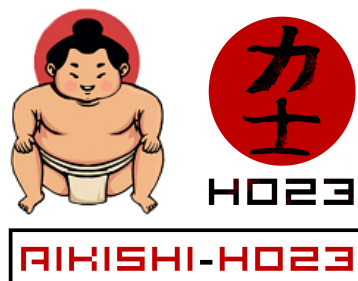
The Sumobot project is a robotic systems project that focuses on designing and implementing a small autonomous robot capable of engaging in sumo-style competitions. The project involves hardware design, software development, and integration of various sensors and actuators.

This manual shows complete information about the functionality, assembly steps, and list of materials of our sumobot named Rikishi-H023, offering an in-depth guide on both its construction and operation. Developed as the final requirement for the subject "Introduction to Robotic Systems," Rikishi-H023 is more than just a project; it's a testament to our dedication, a tangible embodiment of our theoretical knowledge, and a showcase of our practical skills in the fascinating world of robotics.

As you navigate through this document, you'll uncover the innovative design principles, intricate mechanisms, and strategic functionalities that distinguish Rikishi-H023 in the field of robotics. Beyond being a testament to our hard work, this project is positioned as a valuable resource for future students and robotics enthusiasts. By sharing our experiences and insights, we hope to inspire and guide those who follow in our footsteps, fostering a continuous cycle of innovation and learning in the dynamic field of robotic systems.

PROJECT NAME AND INSPIRATION

Rikishi-H023, the chosen name for our project, is inspired by the Japanese term 'Rikishi,' which refers to sumo wrestlers. This name was selected to encapsulate the strength, stability, and strategic prowess that sumo wrestlers embody, traits we aimed to emulate in our robot's design and functionalities. The 'H' in the name stands for 'Host,' indicating the robot's role in hosting and managing various tasks, while '023' signifies its model year, 2023, marking a new era in robotic innovation.



DESIGN

The design of our Sumobot, Rikishi-H023, underwent meticulous refinement to align with the specified requirements and limitations, ensuring fair field in competitions. Among these considerations were the constraints related to mass and dimensions, which were thoughtfully provided to all participants.



Mass Considerations:

Our design team diligently adhered to the prescribed mass limits, recognizing the paramount importance of ensuring fair competition. By staying within these constraints, our goal was to provide each participating Sumobot with an equal opportunity to showcase its capabilities. This commitment to fairness not only upholds the competition's integrity but also highlights our team's dedication to following given guidelines.

The mass of the robot holds significant importance as it directly influences the robot's agility and energy consumption. Striking a balance between robust design and intended applications was crucial for Rikishi-H023. During the official weighing, our robot totaled **1.3 kg**, confirming our meticulous attention to meeting the specified requirements.

Dimensional Constraints:

In addition to mass, meticulous attention was devoted to the specified dimensions outlined for the Sumobot. These constraints served as guiding principles for our design choices, fostering an environment where innovation, strategy, and the efficiency of robotic systems took precedence over external factors. In our specific case, our instructor imposed constraints, capping our measurements at a maximum of 25 cm in both length and height.

The Rikishi-H023 is defined by specific dimensions, covering length, width, and height. These measurements were thoughtfully determined to optimize the robot's functional efficiency and maneuverability. The precise specifications contribute to a well-balanced and stable robot. For Rikishi-H023, the dimensions are as follows:

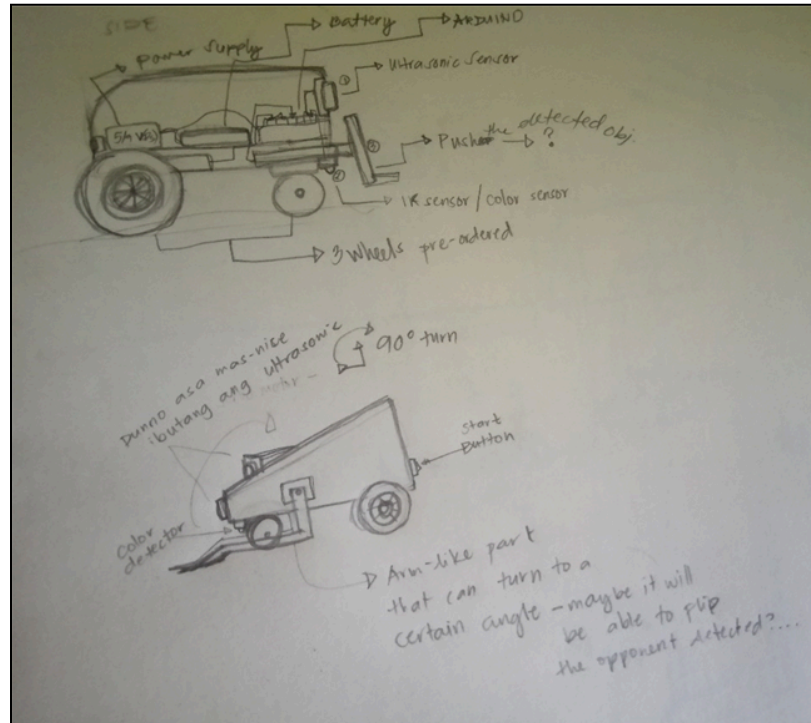
Length = 25 cm; **Width** = 15 cm; **Height** = 17.5 cm.

By tailoring our design to these stringent requirements, Rikishi-H023 not only meets the necessary criteria for participation but also exemplifies our commitment to a fair and competitive arena. This approach not only underscores our technical prowess but also ensures a challenging yet equitable landscape for all participants.

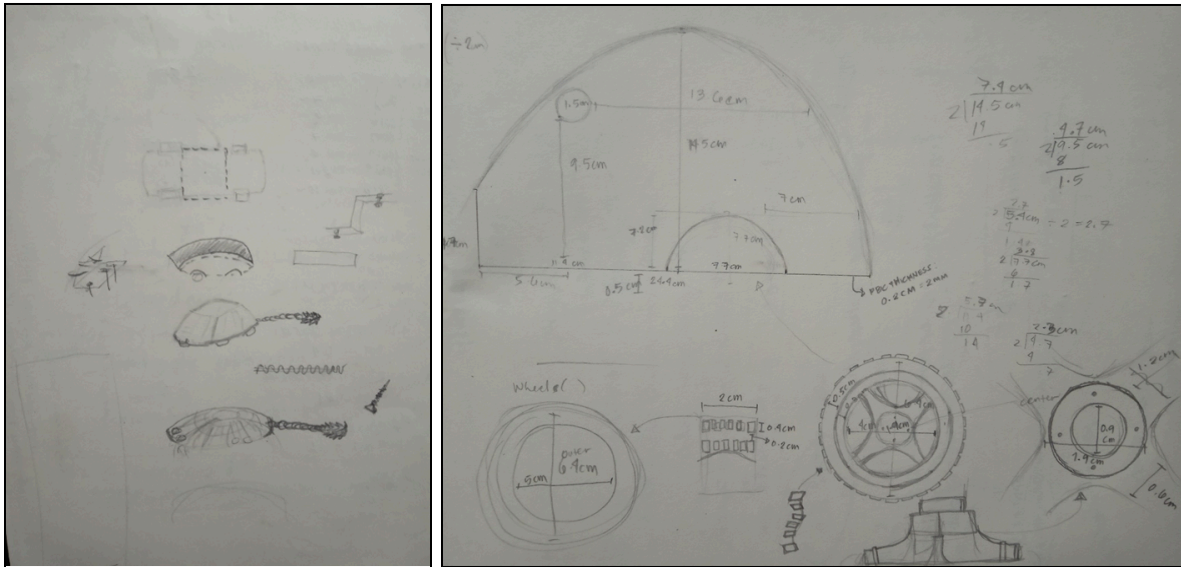
Design Rationale:

The design of Rikishi-H023 was inspired by a combination of practical requirements and innovative engineering concepts. The process involved iterative prototyping, where we continuously refined the robot's design based on trial and error, ensuring each aspect was optimized for both performance and aesthetics.

Initial Design Illustration:



Final Design Illustration:



As shown in the illustrations above, during our initial design phase, we recognized certain drawbacks related to balance if we were to proceed with our original plan, given the use of only three wheels. To address this, we opted for a redesign centered around a three-wheeled car configuration, drawing inspiration from the form of ninja turtles—complete with a design that amusingly echoes their protective shells.

For a more optimized and effective design, we incorporated port holes into the structure and added an emergency double-throw switch. This switch allows a direct connection between the motor and the 20V battery and a connection to the converted 12V supply, enabling us to enforce maximum power when desired or if the battery voltage drops below 16V. This modification addresses the observed decrease in motor power as the battery voltage lowers, preventing performance slowdowns without requiring a full recharge. Additionally, for user convenience, we included charging port holes and a dedicated port for connecting directly to the Arduino Uno. This feature streamlines the programming process without the need to fully open the robot, providing easy access to the Arduino's upload port.

This adjustment not only fortified the defense of our robot's internal components but also significantly improved its balance, minimizing the risk of tipping or imbalance upon impact. The turtle-inspired shell cover played a crucial role in achieving this stability. Moreover, this unique design provided a secure enclosure for our sensors, safeguarding them from potential impacts during battles with other sumobots. In essence, the turtle-inspired design not only enhanced the resilience and balance of our robot but also strategically protected its vital components during intense competition.

MATERIALS/COMPONENTS

The construction of Rikishi-H023 involved a carefully curated selection of materials and components, each chosen for its reliability, functionality, and compatibility with the overall system. The list below provides a detailed overview of these components:

POWER SYSTEM

- **20V Lithium Battery:** A high-capacity battery selected for its long-lasting performance, ensuring prolonged operational periods without the need for frequent recharging.

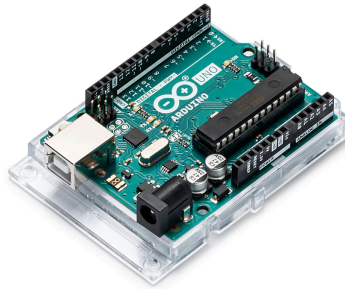


- **DC 24V to DC 12V Step Down / Buck Converter Voltage Regulator:** A crucial component for voltage regulation, protecting sensitive electronic parts from potential damage due to voltage fluctuations.

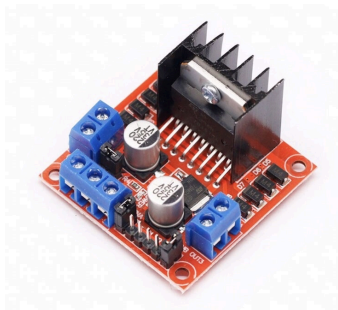


MOTION AND CONTROL

- **Arduino Uno Microcontroller:** The most essential component to build an autonomous system. This enables sensor interpretation, algorithm execution, and precise coordination of movements for our Sumobot.



- **L298n Motor Driver:** A high-efficiency motor driver, central to the precise control of the robot's movement.

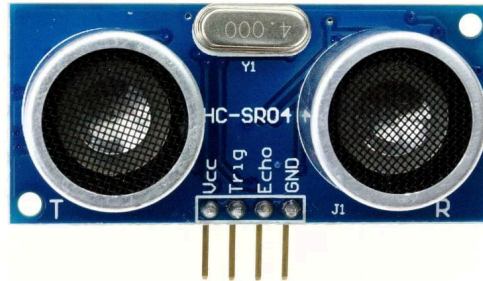


- **Car Chassis Set:** The structural foundation of the robot, designed for durability and flexibility. The set includes 2 DC motors, 2 car wheels, 1 caster wheel, screws, bolts, an acrylic base, and a battery case.

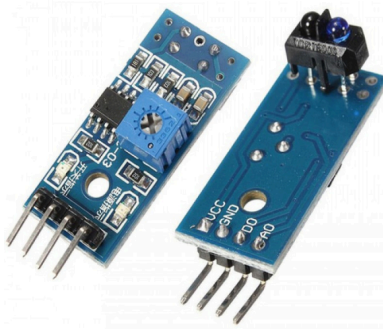


SENSORS

- **Ultrasonic Sensor:** Implements distance sensing technology for navigation and obstacle avoidance.



- **TCRT5000 Color Sensor:** Enhances the robot's ability to differentiate between colors, a feature crucial for certain task-specific applications.



CONNECTIVITY

- **Jumper Wires:** This wire was used for the connection of electronic components such as sensors, resistors, and LEDs to the Arduino Uno microcontroller.
- **Red And Black Electrical Wire:** This wire was utilized as a secure and reliable connection for handling a 24V power supply in our project.
- **Power Jack Connector:** This served as a solderless connection that allowed us to reconnect wiring connections during the troubleshooting phase of our hardware.
- **Universal PCB Board:** This board serves as the primary platform for integrating and soldering electronic elements, ensuring a neat and organized layout.

INDICATORS AND CIRCUIT MANAGEMENT

- **LEDs:** Used for visual signaling, they provide clear operational feedback and status indicators.
- **Resistors:** Essential for controlling and stabilizing the flow of electricity within the robot's circuitry, protecting against current overloads.

OUTER BODY/SHELL

- PBC Pipe
- Screw-bolts
- Sealants
- Epoxy
- Paint
- Sand paper

Each of these components was selected and integrated with the utmost consideration for the overall performance and efficiency of Rikishi-H023, ensuring a harmonious and effective robotic system.

ASSEMBLY DOCUMENTATION

Step 1: Planning and Design

This stage was pivotal in laying the groundwork for Rikishi-H023, with emphasis on ensuring that all components were compatible and met the functional requirements of the robot. Design finalization involved rigorous evaluations and refinements, leading to the determination of the final blueprints for assembly.

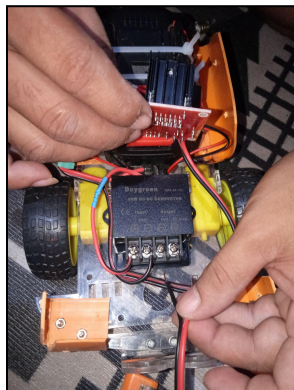
Step 2: Assemble Main Body/Base

The assembly of the main body or base of the robot was the first physical step in the construction process. This involved the integration of the car chassis, wheels, DC motors, battery, and switch.



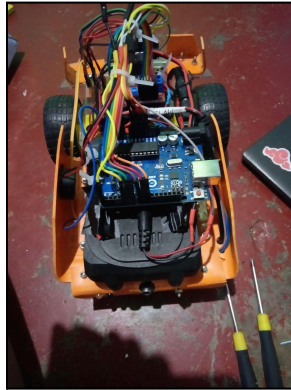
Step 4: Attach Power System

The power system, including the battery, L298n Motor Driver, and 12V converter, was then attached to the chassis. This step was crucial in establishing the robot's power infrastructure, ensuring that all components received the necessary power for operation.



Step 5: Attach Arduino Uno and Wiring Connections

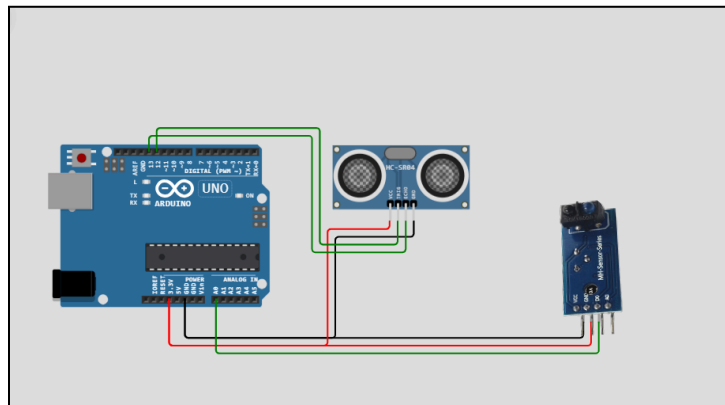
Subsequently, the Arduino Uno was attached, serving as the central processing unit for Rikishi-H023. This step also included the careful arrangement and fixing of wire connections, ensuring a neat and organized layout while maintaining functional integrity.



Step 6: Attach Sensors

The various sensors, including the Ultrasonic Sensor, TCRT5000 Color Sensor, and IR Sensor, were then attached. These sensors are integral to the robot's interactive capabilities, enabling it to navigate, detect objects, and process environmental information.

- Connect TCRT5000 and Ultrasonic Sensor to Arduino uno. See the schematic diagram below.



- **Ultrasonic Sensor Pin Connections:**
 - Trig - Pin D12
 - Echo - Pin D13
- **TCRT5000 Pin Connections:**
 - A0 - Pin A0

Step 7: Create the Sumobot's Cover

The creation of the sumobot's cover involved the use of materials such as PVC Pipe and bolts. This step was not only about protecting the internal components but also about enhancing the robot's aesthetic appeal. The specifics of the materials and design are to be confirmed and elaborated upon.



Step 8: Final Assembly and Design Finalization

The final step in the assembly process was to bring together all components, finalize the design, which included painting and organizing wire connections. A critical reminder at this stage was to meticulously document the pin connections for the sensors, motors, and other components, ensuring ease of troubleshooting and future modifications.



SYSTEM BLOCK DIAGRAM

The system block diagram provides a schematic representation of Rikishi-H023's electronic architecture. At the system's core is the microcontroller, which functions as the central processing unit, orchestrating the robot's operations. Power is supplied by a dedicated source, ensuring consistent energy distribution to the motor driver, which in turn controls the two motors responsible for locomotion. The integration of sensors, including the Line Tracker Sensor and the Ultrasonic Sensor, allows for environmental interaction and navigation, feeding data back to the microcontroller for real-time processing.

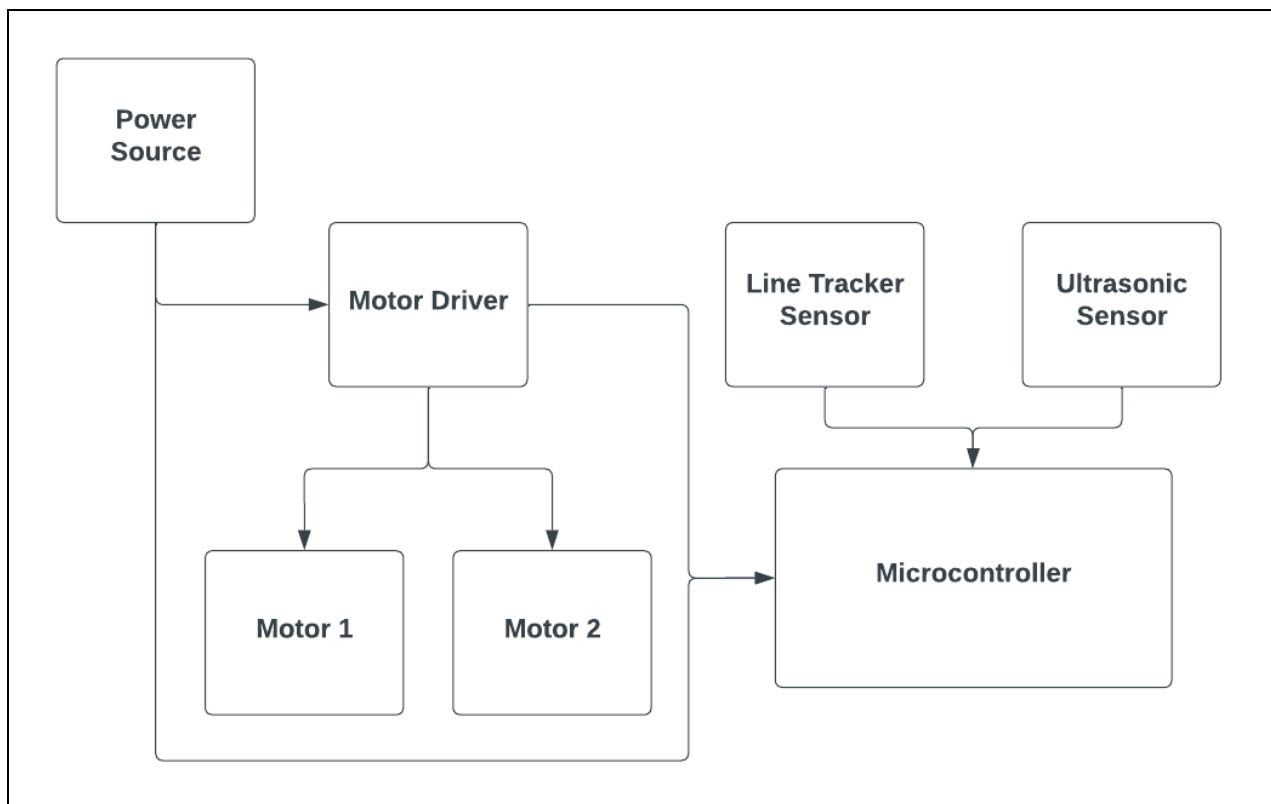


Image 1. Rikishi H023 System Block Diagram

ALGORITHM FLOWCHART

The algorithm flowchart delineates the logic sequence employed by Rikishi-H023's microcontroller. Upon initiation, the robot awaits user input to activate the power system. Post activation, the robot enters a scanning mode, utilizing the Ultrasonic Sensor to detect the proximity of objects within its environment. The Line Tracker Sensor is engaged to ensure the robot remains within its operational boundaries. If an enemy or obstacle is detected within a predefined distance, the robot engages in a strategic maneuver to approach and exert a pushing force. This algorithmic process is crucial for Rikishi-H023's functionality in navigational and competitive contexts.

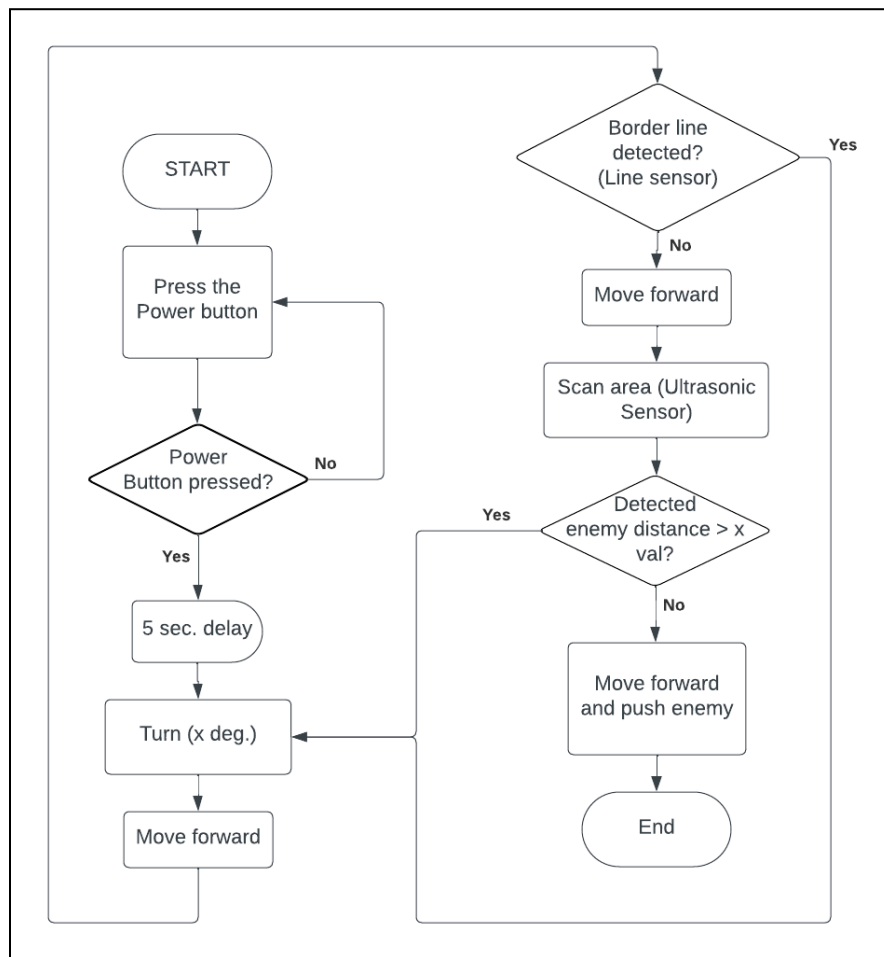


Image 2. Rikishi H023 Algorithm Flowchart

STRATEGY

These are the strategies implemented during battles. In Figures 2.1 and 2.2, we illustrate the attack strategy programmed in our robot. Initially, it scans for the enemy using the ultrasonic sensor, and upon detecting the enemy within a predetermined distance, the robot engages by pushing it off until it reaches the out zone.

In Figures 2.3 and 2.4, we showcase the strategy implemented when the robot senses that it is nearing the boundaries. It promptly initiates a turn to ensure it stays within the designated boundary limits.

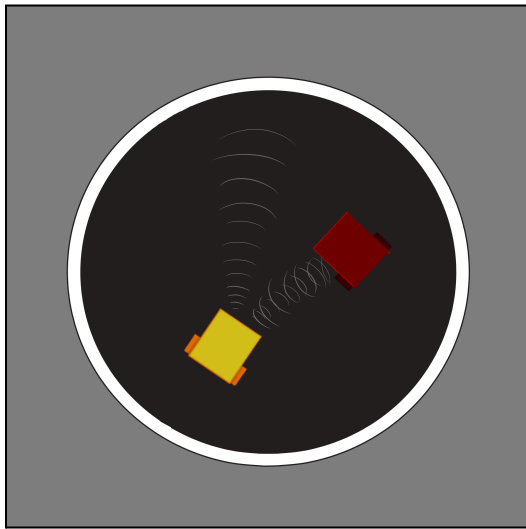


Figure 2.1 Scanning

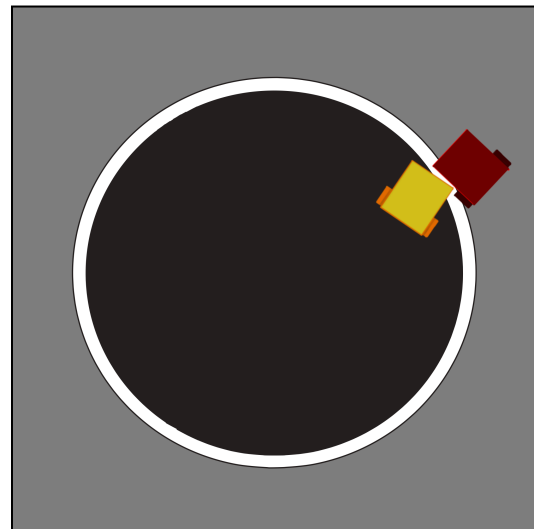


Figure 2.2 Pushing

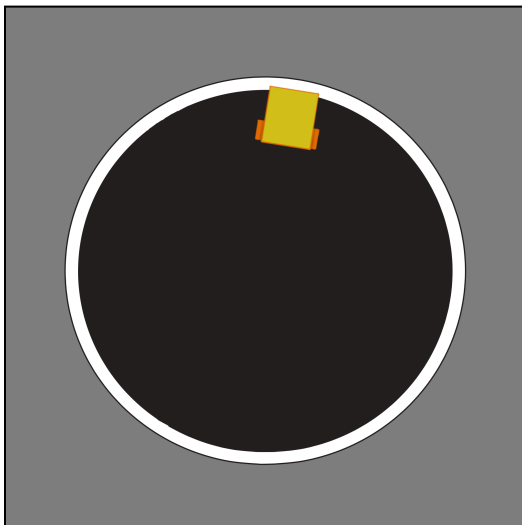


Figure 2.3 White line detection

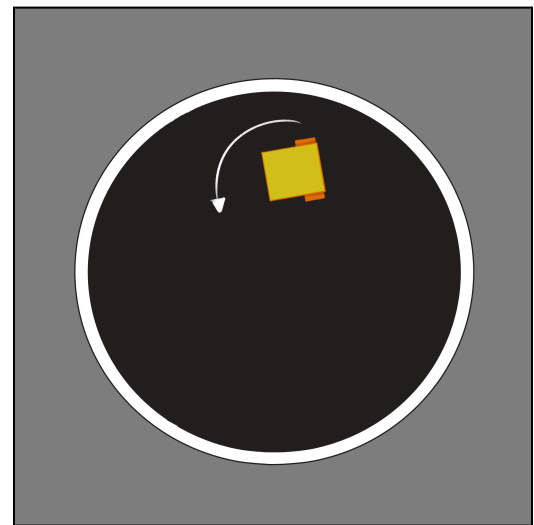
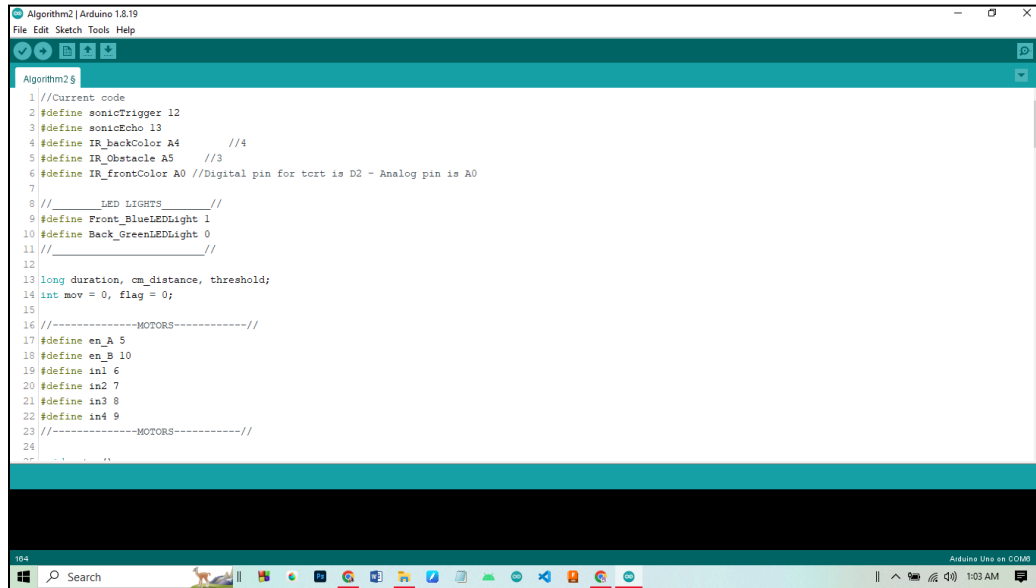


Figure 2.4 Turn back inside

SOFTWARE

To make the sumobot an autonomous robot, we used Arduino IDE to write the program with C++ programming language along with the logic formulated through our algorithm flowchart.



Program Code:

```
#define sonicTrigger 12

#define sonicEcho 13

#define IR_backColor A4 //4

#define IR_Obstacle A5 //3

#define IR_frontColor A0 //Digital pin for tcert is D2 - Analog pin is A0

//_____LED LIGHTS_____//

#define Front_BlueLEDDLight 1

#define Back_GreenLEDDLight 0

//_____//

long duration, cm_distance, threshold;

int mov = 0, flag = 0;
```

```

//-----MOTORS-----//

#define en_A 5

#define en_B 10

#define in1 6

#define in2 7

#define in3 8

#define in4 9

//-----MOTORS-----//

void setup(){

    delay(3000);

    // put your setup code here, to run once:

    for(byte x = 5; x < 11; x++){

        pinMode(x, OUTPUT);

    }

    pinMode(12, OUTPUT);

    pinMode(A0, INPUT);

    pinMode(4, INPUT);

    pinMode(3, INPUT);

    pinMode(13, INPUT);

    pinMode(A1, OUTPUT);

    pinMode(A2, OUTPUT);

    Serial.begin(9600);

}

class Rikishi_Func{

```

public:

```
void forward(){  
    digitalWrite(in1, LOW);  
    digitalWrite(in2, HIGH);  
    digitalWrite(in3, LOW);  
    digitalWrite(in4, HIGH);};
```

```
void backward(){  
    digitalWrite(in1, HIGH);  
    digitalWrite(in2, LOW);  
    digitalWrite(in3, HIGH);  
    digitalWrite(in4, LOW);};
```

```
void turnLeft(){  
    digitalWrite(in1, LOW);  
    digitalWrite(in2, HIGH);  
    digitalWrite(in3, HIGH);  
    digitalWrite(in4, LOW);};
```

```
void turnRight(){  
    digitalWrite(in1, HIGH);  
    digitalWrite(in2, LOW);  
    digitalWrite(in3, LOW);  
    digitalWrite(in4, HIGH);};
```

```
void halt(){  
    digitalWrite(in1, LOW);  
    digitalWrite(in2, LOW);  
    digitalWrite(in3, LOW);
```

```

    digitalWrite(in4, LOW);
}

void setSpeedMotor_A(byte x){
    analogWrite(en_A, x);
}

void setSpeedMotor_B(byte x){
    analogWrite(en_B, x);
}

void FrontSign_Blink(){
    analogWrite(Front_BlueLEDLight, 255);
    delay(1000);
    analogWrite(Front_BlueLEDLight, 0);
    delay(1000);
}

void BackSign_Blink(){
    analogWrite(Front_BlueLEDLight, 255);
    delay(1000);
    analogWrite(Front_BlueLEDLight, 0);
    delay(1000);
}

void All_LEDLight(){
    analogWrite(Back_GreenLEDLight, 255);
    analogWrite(Front_BlueLEDLight, 255);
}

void Off_LED(){

```



```

    analogWrite(Back_GreenLEDLight, 0);
    analogWrite(Front_BlueLEDLight, 0);
}

void distance_Calculation(){
    digitalWrite(sonicTrigger, LOW);
    delayMicroseconds(2);
    digitalWrite(sonicTrigger, HIGH); // Transmit Waves For 10us
    delayMicroseconds(10);
    duration = pulseIn(sonicEcho, HIGH); // Receive Reflected Waves
    cm_distance = duration / 58.2; // Get Distance in centimeters
}

};

Rikishi_Func Start;

void loop(){
    Start.setSpeedMotor_A(150);
    Start.setSpeedMotor_B(150);
    Start.distance_Calculation();
    if( analogRead(IR_frontColor) < 90){
        Start.halt();
        Start.backward();
        delay(100);
    }
    while( analogRead(IR_frontColor) > 90){
        Start.distance_Calculation();
        if (cm_distance > 40){

```

```

        Start.turnRight();
        delay(200);
        Start.halt();
        delay(1000);
        Serial.println("Enemy too far");
    }

    if (cm_distance < 40){
        Start.forward();
        delay(500);
        Start.halt();
        delay(100);

        if (cm_distance < 20);
        Start.backward();
        delay(150); //100
        Start.forward();
        delay(500);
        Start.halt();
        delay(500);

        Start.All_LEDLight();
        Start.Off_LED();
    }
}

```

```
Serial.println(analogRead(IR_frontColor));  
Start.distance_Calculation();  
    Serial.print("DISTANCE: ");  
    Serial.print(cm_distance);  
    Serial.println(" cm");  
Serial.println(analogRead(IR_backColor));  
    Serial.print("IR BACK");  
    delay(100);  
}
```

BATTLE DOCUMENTATION



Image 6. 1bot VS. Rikishi-H023 Battle

OUTPUT

The output of Rikishi-H023 is multifaceted, reflecting the sum of its design, components, and programming. Primarily, the robot demonstrates a high degree of responsiveness and precision in navigation tasks, attributable to its advanced sensor array and well-calibrated motor system. In competitive scenarios, the robot's output is measured by its ability to stay within the arena while effectively locating and pushing opponents out of bounds.

PERFORMANCE

Rikishi-H023's performance has been evaluated across a series of metrics, including maneuverability, speed, stability, and sensor accuracy. Through rigorous testing, the robot has shown commendable agility and consistency in executing programmed tasks. One notable aspect of Rikishi-H023's performance is the controllable speed achieved by adjusting the PWM values. This feature allows for fine-tuned speed management, offering flexibility in different operational scenarios. Its robust construction and strategic design contribute to a performance that meets the objectives set out at the inception of the project.

DRAWBACKS

Despite its successes, Rikishi-H023 has certain limitations. One such drawback is the finite battery life, which constrains the robot's operational duration and necessitates frequent recharging intervals. Additionally, the weight distribution of the robot has occasionally affected its balance, a factor that could be critical in competition settings.

1. Weight Distribution and Balance:

- The weight distribution of the robot has, at times, impacted its balance.
- This drawback becomes particularly critical in competition settings where stability is crucial.

2. Line Detection Program Performance:

- The line detection program, triggered by the TCRT5000 Sensor, exhibited delays.
- Immediate re-entry into the arena upon detecting the white line resulted in inaccuracies.
- The delays compromised the accuracy of the detection system when the robot left the arena.

3. Scanning Strategy Limitation:

- The initial strategy involved scanning the area to detect enemies using the ultrasonic sensor.
- This approach proved to be slow, and there were instances where the robot was detected by the enemy first.
- The scan-setting, involving slow turns, contributed to the delay and potential detection by adversaries.

RECOMMENDATIONS

To enhance Rikishi-H023's functionality, several recommendations are proposed.

1. Adaptive Weight Distribution Techniques:

Proposal: Implement adaptive weight distribution techniques.

Rationale: Enhance stability and agility, especially in dynamic environments where balance is crucial. This improvement could positively impact Rikishi-H023's performance during sumo-style competitions.

Implementation: Investigate and incorporate mechanisms for dynamically adjusting the robot's weight distribution based on its current state and environmental conditions.

2. Color Detection Sensor Optimization:

Proposal: Ensure the color detection sensor is active and responsive without delays.

Rationale: In the programming and strategy execution, the reliability of the color detection sensor is crucial for accurate decision-making. Any delays or issues in its functionality can impact the robot's performance.

Implementation: Conduct thorough testing, adjust sensor placements if necessary, and optimize the program to eliminate delays, possibly through a trial-and-error method to fine-tune the color detection process.

3. Sensor Technology Upgrade:

Proposal: Consider upgrading sensor technology.

Rationale: Upgrading sensors can refine Rikishi-H023's environmental interaction capabilities, providing more accurate and comprehensive data for decision-making.

Implementation: Explore the latest sensor technologies relevant to the project requirements and integrate upgraded sensors to enhance the robot's overall sensory perception.

These recommendations, when implemented, aim to elevate Rikishi-H023's performance, responsiveness, and adaptability, contributing to its success in robotic competitions.

CONCLUSION

In conclusion, Rikishi-H023 represents a significant step forward in the application of robotic systems within competitive arenas. It encapsulates the theoretical knowledge and practical engineering skills acquired through our coursework, and its development has provided valuable insights into robotic design and operation. While there are areas for improvement, the project has met its foundational objectives and established a solid framework for future innovation and development in the field of robotics.

Engaging in hands-on activities like building Rikishi-H023 proves instrumental in helping students envision and apply theoretical concepts learned in their coursework. Such projects serve as effective methods for students to gain a comprehensive understanding of the fundamentals of robotics, paving the way for future exploration and innovation. There exists a plethora of similarly engaging projects that can foster learning while providing an enjoyable experience, making robotics education an exciting and educational endeavor.